Week 5 · Part E — justification of the three health figures

Qurat ul Aien · Roll No. AI-016 · CDB601220 Data Visualization · Week 5 · synthetic course data; geolib/vizlib used unchanged, no geopandas

1 · Which variable got which channel, and why (slide 8's budget)

D1 spends the channels in the order the reader must compare: the two quantities to read precisely get the two **positions** — x = GDP per capita on a **log** axis (a multiplicative spread, $283–$120,219) and y = healthcare access (a bounded 0–100 index). Population gets **area**, which is read only roughly, so it carries the 'how many people' context and a size legend with real values (1, 50, 350 M) lets readers convert it back; the radius goes as √, never linearly (sizes = 16–1,600 pt², 10× across — Part B1). Region is identity, not amount, so it gets **hue** (six Okabe–Ito colours) plus the region letter on each mark, because the green/pink pair falls to ΔE 7.5 under deuteranopia. The GDP outlier (Aldonia, 9.2× the next) is kept and labelled; the log axis holds it.

2 · Parallel coordinates decisions

Scaling: percentile rank, chosen before drawing, because min–max puts 58/60 countries under 0.10 on GDP and z-scoring puts Aldonia at +7.5 sd (Part B3); rank keeps each country's order, which is what crossings are read from, and I printed the real min and max at each axis end. **Order: the greedy |r| chain** (access → cases → urban → median age → GDP), computed with a pandas-3-safe copy of geolib.corr_chain (the original crashes on pandas 3 — reported as a finding). **Clutter:** all 60 grey, Borealis brushed. It showed an X between access and cases (81% of lines cross, r = −0.80) and — the relationship a Pearson scatter-matrix would have hidden — median age and GDP running almost parallel: **ρ = +0.87 but Pearson r = +0.28**, because one outlier drags r down; only 16% of lines cross in that gap. PCA confirms two axes miss 16.3% of the variation.

3 · Spatial decisions

Denominator: resident population, so the map shows cases per 100,000 residents (written in the legend title) — the question is risk, and counts track population (r = +0.986). **Classification:** Jenks natural breaks, 5 classes (GVF 0.969 vs 0.908 for quantile), breaks printed in the legend with the number of districts in each. **Projection:** Lambert cylindrical equal-area at 30.5°N, because colour over area makes area the channel; Mercator would have inflated the north 1.49× and the south 1.24× (Part B4 — geolib's self-check figure uses the linear, not the area, scale). The join was checked: 24/24 matched, 0 failed. **Suppression:** rule set at < 20 cases per district; none qualified (smallest 186 cases), so nothing was merged. For the small-area file the same rule would merge every unit under ~43,000 people, the population below which 9 and 18 per 100,000 cannot be told apart (Part B5) — at the price of losing U32's signal.

4 · The counterfactual

A raw-count map of this outbreak would have coloured **D17 Qilabad, D13 Mehrabad and D09 Ilyaspur** darkest — the three biggest urban districts, 6.3–11.0 M people each — and told a health ministry to send its teams to the cities. Two of the three run at or below the national rate (51.0 and 49.9 vs 52.8). The districts actually at highest risk, **D19, D20 and D21** in the north (69–76 per 100,000, 1.1–1.8 M people), would have been pale. The cost lands on two groups: the city residents whose districts are labelled 'outbreak areas' every week, and the northern residents whose real risk is never shown. **[Part C readers not yet run — after `python partC_maps.py --record` for each of at least five real readers, re-run build_reports.py and this sentence is replaced by the measured mean overlap.]** Week 4's lesson applies unchanged: the denominator is a modelling decision, and a map is never neutral because it points at a place where people live.